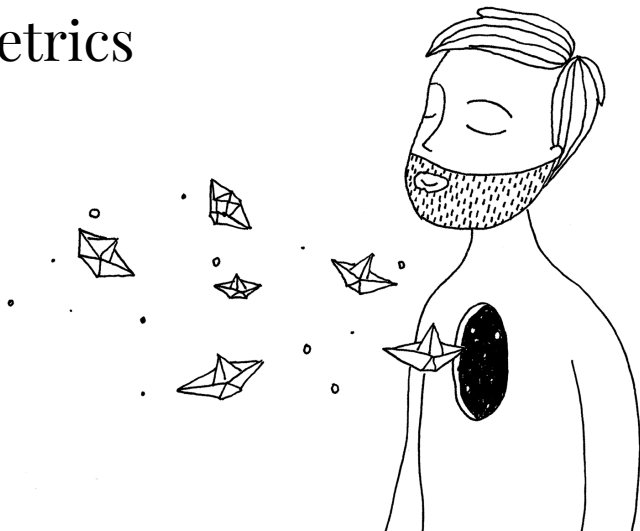


Nova SBE Econometrics

Classes 6 & 7
Dummy Variables
Instrumental Variables

PAULO FAGANDINI

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Nova SBE Econometrics

Dummy Variables & Qualitative Information

Debrief: Why Dummy Variables?

Key idea. Many economic variables are *qualitative*: gender, marital status, industry, region, position played.

We encode them as binary (0/1) **dummy variables** and include them in OLS regressions.

Interpretation rule

The coefficient on a dummy measures the *ceteris-paribus difference in the conditional mean* of y between that group and the **reference (base) category**.

Dummy variable trap. With m mutually exclusive categories, include at most $m - 1$ dummies alongside an intercept. Including all m creates *perfect multicollinearity*.

Debrief: Polynomial Terms & Interactions

Quadratic model:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots, \quad \frac{\partial y}{\partial x} = \beta_1 + 2\beta_2 x.$$

Turning point (optimum): $x^* = -\hat{\beta}_1 / (2\hat{\beta}_2)$.

Interaction terms:

$$y = \beta_0 + \beta_1 x + \beta_2 D + \beta_3 (D \cdot x) + u, \quad \frac{\partial y}{\partial x} = \beta_1 + \beta_3 D.$$

F-test for joint significance (q restrictions):

$$F = \frac{(R_{ur}^2 - R_r^2)/q}{(1 - R_{ur}^2)/(n - k - 1)} \sim F(q, n - k - 1).$$

Debrief: Individual Significance (t-test)

$$t = \frac{\hat{\beta}_j}{\widehat{\text{se}}(\hat{\beta}_j)} \sim t_{n-k-1}.$$

- Two-tailed ($H_a: \beta_j \neq 0$): reject if $|t| > t_c$.
- One-tailed ($H_a: \beta_j > 0$): reject if $t > t_c$.

Large- n : $t_c \approx 1.96$ (two-tailed, 5%) and $t_c \approx 1.645$ (one-tailed, 5%).

Testing a linear combination $H_0 : \beta_j + \beta_k = 0$

$$t = \frac{\hat{\beta}_j + \hat{\beta}_k}{\sqrt{\widehat{\text{V}}(\hat{\beta}_j) + \widehat{\text{V}}(\hat{\beta}_k) + 2 \widehat{\text{Cov}}(\hat{\beta}_j, \hat{\beta}_k)}}$$

Exercise 5.5 — Basketball Players: Data

Variables:

- `points` — points scored per game
- `exper` — years as a professional player
- `guard`, `forward` — position dummies (reference: *center*)

Estimated equation ($n = 269$, $R^2 = 0.091$, $\bar{R}^2 = 0.077$):

$$\widehat{\text{points}} = \underset{(1.18)}{4.76} + \underset{(0.33)}{1.28 \text{ exper}} - \underset{(0.024)}{0.072 \text{ exper}^2} + \underset{(1.00)}{2.31 \text{ guard}} + \underset{(1.00)}{1.54 \text{ forward}}$$

Exercise 5.5 – Questions

skip to Exercise 5.9 →

- (a) Interpret the regression coefficients.
- (b) Why can't we include a center dummy?
- (c) Holding experience fixed, does a guard score more than a center? Is the difference statistically significant?
- (d) Using the restricted model below, test whether experience influences points.
- (e) Extended model with `marr`: do married players score more points?
- (f) Model with marital-status interactions (see table): is marital status jointly significant?
- (g) From the interaction model, what is the optimal number of years of experience for married and unmarried players?

Restricted model for (d) ($n = 269$, $R^2 = 0.020$):

$$\widehat{\text{points}} = 8.52 + 2.40 \text{ guard} + 1.68 \text{ forward}$$

(0.86) (1.03) (1.03)

Exercise 5.5(a) — Interpreting Coefficients

Marginal effect of experience:

$$\frac{\partial \widehat{\text{points}}}{\partial \text{exper}} = 1.28 - 2 \times 0.072 \times \text{exper} = 1.28 - 0.144 \text{exper}.$$

- Effect is *positive but decreasing*.
- Turning point: $\text{exper}^* = 1.28/0.144 \approx 8.9$ years.
- 1st year: +1.28 pts; 5th year: $1.28 - 0.144 \times 4 = 0.70$ pts.

Position dummies (reference = center):

- Guards score **2.31 more points** per game than centers, *c.p.*
- Forwards score **1.54 more points** per game than centers, *c.p.*

← back to problem

Exercise 5.5(b) — Dummy Variable Trap

Every player occupies *exactly one* position:

$$\text{guard} + \text{forward} + \text{center} = 1 \quad \forall i.$$

Adding center creates a perfect linear combination with the intercept column:

$$\text{center} = 1 - \text{guard} - \text{forward}.$$

⇒ **Perfect multicollinearity**: $(X'X)$ is singular and OLS is undefined.

Rule

With m mutually exclusive categories, include at most $m - 1$ dummies and an intercept. The omitted category is the **reference group**.

Alternative: drop the intercept and include all m dummies, but group comparisons are less straightforward.

← back to problem

Exercise 5.5(c) — Guards vs. Centers

$$H_0 : \beta_3 = 0 \text{ vs. } H_a : \beta_3 > 0$$

$$t = \frac{2.31}{1.00} = 2.31 \quad t_c \approx 1.645 \text{ (one-tailed, 5\%, } df = 264)$$

Since $2.31 > 1.645 \Rightarrow$ **reject** H_0 .

Conclusion

Statistically significant evidence that guards score more points per game than centers, holding experience fixed.

[← back to problem](#)

Exercise 5.5(d) — Is Experience Significant?

$H_0 : \beta_1 = \beta_2 = 0$ vs. $H_a : \text{not } H_0$

F-test ($q = 2$, $R_{ur}^2 = 0.091$, $R_r^2 = 0.020$):

$$F = \frac{(0.091 - 0.020)/2}{(1 - 0.091)/(269 - 4 - 1)} = \frac{0.0355}{0.003443} = 10.31$$

$F_{0.05}(2, 264) \approx 3.00$. Since $10.31 \gg 3.00 \Rightarrow$ **reject** H_0 .

Conclusion

Strong evidence that experience is jointly significant in explaining points scored per game.

[← back to problem](#)

Exercise 5.5(e) — Marital Status (simple)

Extended model:

$$\widehat{\text{points}} = 4.70 + 1.23 \text{ exper} - 0.07 \text{ exper}^2 + 2.29 \text{ guard} + 1.54 \text{ forward} + \underset{(0.74)}{0.58 \text{ marr}}$$

$$H_0 : \beta_5 = 0 \text{ vs. } H_a : \beta_5 > 0$$

$$t = \frac{0.58}{0.74} = 0.78 < 1.645 \Rightarrow \text{fail to reject } H_0.$$

Conclusion

No statistically significant evidence that married players score more points, holding position and experience fixed.

← back to problem

Exercise 5.5(f) — Interaction Model & F-test

	Coef.	Std. Err.	t	$P > t $
exper	0.703	0.434	1.62	0.107
expersq	-0.030	0.033	-0.90	0.367
guard	2.251	1.000	2.25	0.025
forward	1.629	1.002	1.63	0.105
marr	-2.538	2.038	-1.24	0.214
marr_exper	1.280	0.682	1.88	0.062
marr_expersq	-0.094	0.049	-1.92	0.057
Constant	5.816	1.349	4.31	0.000

$n = 269, R^2 = 0.1058, \bar{R}^2 = 0.0818, F(7, 261) = 4.41$

$H_0 : \beta_5 = \beta_6 = \beta_7 = 0$ vs. $H_a : \text{not } H_0$

$$F = \frac{(0.1058 - 0.091)/3}{(1 - 0.1058)/(269 - 7 - 1)} = 1.44 < F_{0.05}(3, 261) \approx 2.64 \Rightarrow \text{fail to reject } H_0.$$

← back to problem

Exercise 5.5(g) — Optimal Experience

$$\frac{\partial \widehat{\text{points}}}{\partial \text{exper}} = (0.703 - 0.060 \text{ exper}) + \text{marr} (1.280 - 0.188 \text{ exper})$$

Unmarried (marr=0):

$$0.703 - 0.060 \text{ exper}^* = 0 \Rightarrow \text{exper}^* \approx 11.7 \text{ years}$$

Married (marr=1):

$$1.983 - 0.248 \text{ exper}^* = 0 \Rightarrow \text{exper}^* \approx 8.0 \text{ years}$$

Takeaway

Married players peak earlier (≈ 8 yrs) than unmarried players (≈ 12 yrs) — recall though that marital status is jointly insignificant.

[← back to problem](#)

Exercise 5.9 — SAT Scores: Data

Variables: sat (combined SAT score), hsize (graduating class size, in hundreds), female, black

Estimated equation ($n = 4,137$, $R^2 = 0.0858$):

$$\widehat{\text{sat}} = 1028.10 + 19.30 \text{ hsize} - 2.19 \text{ hsize}^2 - 45.09 \text{ female} - 169.81 \text{ black} + 62.31 \text{ female} \times \text{black}$$

(6.29)(3.83)(0.53)(4.29)(12.71)(18.15)

Reference group: *nonblack males* (female=0, black=0).

Exercise 5.9 — Questions

skip to Section 2: Instrumental Variables →

- (a) Is there strong evidence that hsize^2 should be included? What is the optimal high-school size?
- (b) Holding hsize fixed, what is the estimated difference in SAT scores between nonblack females and nonblack males? How significant is it?
- (c) What is the estimated difference between nonblack males and black males? Test whether the difference is zero.
- (d) What is the estimated difference between black females and nonblack females? What would you need to test its significance?

Exercise 5.9(a) — Quadratic Term & Optimal Size

$H_0 : \beta_2 = 0$ vs. $H_a : \beta_2 \neq 0$:

$$t = \frac{-2.19}{0.53} = -4.13, \quad | -4.13 | > 1.96 \Rightarrow \text{reject } H_0.$$

Optimal class size:

$$19.30 - 2 \times 2.19 \text{hsize}^* = 0 \Rightarrow \text{hsize}^* = \frac{19.30}{4.38} = 4.41$$

Interpretation

The optimal graduating class size is $4.41 \times 100 = \mathbf{441}$ students. Classes smaller or larger are associated with lower average SAT scores.

[← back to problem](#)

Exercise 5.9(b) — Nonblack Females vs. Males

$$E[\text{sat} \mid f = 1, b = 0] = \hat{\beta}_0 + \hat{\beta}_3$$

$$E[\text{sat} \mid f = 0, b = 0] = \hat{\beta}_0$$

Difference: $\hat{\beta}_3 = -45.09$

$H_0 : \beta_3 = 0$ vs. $H_a : \beta_3 \neq 0$:

$$t = \frac{-45.09}{4.29} = -10.51, \quad | -10.51 | \gg 1.96 \Rightarrow \text{reject } H_0.$$

Conclusion

Nonblack females score on average **45.09 points less** than nonblack males; the difference is highly significant.

[← back to problem](#)

Exercise 5.9(c) — Black Males vs. Nonblack Males

$$E[\text{sat} \mid f = 0, b = 1] = \hat{\beta}_0 + \hat{\beta}_4$$

$$E[\text{sat} \mid f = 0, b = 0] = \hat{\beta}_0$$

Difference: $\hat{\beta}_4 = -169.81$

$H_0 : \beta_4 = 0$ vs. $H_a : \beta_4 \neq 0$:

$$t = \frac{-169.81}{12.71} = -13.36, \quad | -13.36 | \gg 1.96 \Rightarrow \text{reject } H_0.$$

Conclusion

Black males score on average **169.81 points less** than nonblack males; the difference is highly significant.

← back to problem

Exercise 5.9(d) — Black Females vs. Nonblack Females

$$E[\text{sat} \mid f = 1, b = 1] = \hat{\beta}_0 + \hat{\beta}_3 + \hat{\beta}_4 + \hat{\beta}_5$$

$$E[\text{sat} \mid f = 1, b = 0] = \hat{\beta}_0 + \hat{\beta}_3$$

Difference: $\hat{\beta}_4 + \hat{\beta}_5 = -169.81 + 62.31 = -107.50$

$H_0 : \beta_4 + \beta_5 = 0$ vs. $H_a : \text{not } H_0$

$$t = \frac{\hat{\beta}_4 + \hat{\beta}_5}{\sqrt{\widehat{V}(\hat{\beta}_4) + \widehat{V}(\hat{\beta}_5) + 2\widehat{\text{Cov}}(\hat{\beta}_4, \hat{\beta}_5)}}$$

Alternative

Re-parameterise: let $\theta = \beta_4 + \beta_5$. Re-run with `black` coding the gap between black and nonblack females directly $\Rightarrow$ standard error available from the regression.

[← back to problem](#)

Nova SBE Econometrics

Instrumental Variable Estimation



Debrief: The Endogeneity Problem

OLS requires $E[u | X] = 0$. **Endogeneity** arises from:

- **Omitted variable bias** — confounder z affects both X and y .
- **Simultaneity** — y and X are jointly determined (e.g. supply & demand).
- **Measurement error** — X observed with noise.

Consequence

OLS is *biased and inconsistent* when $\text{Cov}(X, u) \neq 0$. The direction of bias depends on the sign of that covariance.

Debrief: Instrumental Variables & 2SLS

Valid instrument Z for endogenous X :

1. **Relevance:** $\text{Cov}(Z, X) \neq 0$ *(testable: first-stage F)*
2. **Exogeneity:** $\text{Cov}(Z, u) = 0$ *(not directly testable with one instrument)*

Two-Stage Least Squares (2SLS):

Stage 1 Regress X on Z (and controls) $\rightarrow$ obtain $\hat{X}$.

Stage 2 Regress y on $\hat{X}$ (and controls).

2SLS is consistent under relevance + exogeneity, but less efficient than OLS when OLS is also consistent.

Debrief: Weak Instruments

Weak instrument: Z explains very little variation in X .

Consequences: 2SLS biased toward OLS; standard errors unreliable; non-normal distribution even in large samples.

Rule of thumb (Stock–Yogo): $F_{\text{first stage}} > 10$.

Bottom line

A strong, exogenous instrument is both *necessary and sufficient* for 2SLS to deliver credible causal estimates.

Exercise 6.1 — Cigarette Demand: Data

Model: $\ln Q_i = \beta_0 + \beta_1 \ln P_i + u_i$

- $\ln Q_i$: log packs per capita sold in state i (1995)
- $\ln P_i$: log after-tax average real price per pack
- 48 continental US states
- **Instrument:** $Z_i = \ln(\text{SalesTax}_i)$ — portion of tax from the *general* sales tax

	(1) OLS $\ln Q$	(2) First Stage $\ln P$	(3) 2SLS $\ln Q$
$\ln P$	-1.213*** (0.216)		
salestax		0.031*** (0.005)	
$\widehat{\ln P}$			-1.084*** (0.377)
Constant	10.339***	4.617***	9.720***
R^2	0.406	0.471	0.152
F	31.41	40.96	8.28

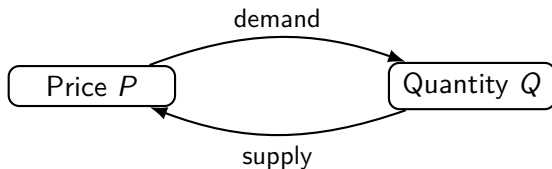
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Exercise 6.1 — Questions

- (a) Is it reasonable to use OLS to estimate the price elasticity of demand? Why?
- (b) Discuss the two conditions that $\ln(\text{SalesTax})$ must satisfy to be a valid instrument.
- (c) How much of the variation in $\ln P$ is explained by the instrument? Relate this to the concept of a weak instrument.
- (d) Explain the difference between the second-stage regression (3) and the OLS regression (1).
- (e) Interpret the coefficient in (3). Compare with (1) and comment.
- (f) What are the main limitations of these models?

Exercise 6.1(a) — Can We Use OLS?

Problem: Simultaneous causality.



Price affects Q demanded, but Q sold also affects equilibrium P . $\Rightarrow \text{Cov}(\ln P, u) \neq 0$
 $\Rightarrow$ OLS is **biased and inconsistent**.

Solution

Use an instrument that shifts the supply curve (hence price) but is uncorrelated with demand shocks.

[← back to problem](#)

Exercise 6.1(b) — Instrument Validity

1. Relevance — $\text{Cov}(Z, \ln P) \neq 0$

The general sales tax is a direct component of the after-tax price per pack: higher tax $\Rightarrow$ higher price.

Testable via first-stage F-statistic.

2. Exogeneity — $\text{Cov}(Z, u) = 0$

The general sales tax applies to all goods and is set by state fiscal policy — not by cigarette-specific demand shocks u_i .

Plausible, but not directly testable with a single instrument.

[← back to problem](#)

Exercise 6.1(c) — Weak Instrument Check

From the first-stage regression (column 2):

$$R^2_{\text{first stage}} = 0.471, \quad F_{\text{first stage}} = 40.96.$$

- The sales tax explains $\approx 47\%$ of the variation in after-tax prices across states.
- $F = 40.96 \gg 10$ — strong evidence *against* a weak instrument.

Reminder

A weak instrument causes 2SLS to be nearly as biased as OLS and renders standard inference unreliable.

[← back to problem](#)

Exercise 6.1(d) — Second Stage vs. OLS

Column (1) — OLS: regresses $\ln Q$ on observed $\ln P$, which contains both demand-driven and supply-driven price variation $\Rightarrow$ endogeneity bias.

Column (3) — 2SLS second stage: regresses $\ln Q$ on *fitted* $\widehat{\ln P}$ from Stage 1.

Key insight

$\widehat{\ln P}$ retains only the exogenous (sales-tax-driven) variation in price, purging the component correlated with u . This isolates the causal demand response.

Note: the second-stage R^2 (0.152) is lower than OLS R^2 (0.406) because only part of the price variation is being used.

[← back to problem](#)

Exercise 6.1(e) — Interpreting the 2SLS Coefficient

2SLS: $\hat{\beta}_1^{IV} = -1.084$

Interpretation

A 1% increase in the real price of cigarettes reduces per-capita consumption by $\approx 1.08\%$, *c.p.* Demand is **elastic** ($|\varepsilon| > 1$).

Comparison with OLS ($\hat{\beta}_1^{OLS} = -1.213$):

- OLS overstates the elasticity (in abs. value) due to simultaneity bias.
- 2SLS reduces the estimate from -1.213 to -1.084 : removing endogenous price variation yields a less elastic demand.
- Both suggest fairly elastic demand — somewhat surprising for an addictive good.

[← back to problem](#)

Exercise 6.1(f) — Limitations

1. **Omitted variables.** State income affects demand *and* correlates with the sales tax $\Rightarrow$ remaining omitted-variable bias.
2. **Single instrument.** Exogeneity not testable with one instrument; a second would allow an over-identification (Sargan/Hansen J) test.
3. **Cross-section only (1995).** Panel data with state fixed effects would absorb time-invariant confounders.
4. **Homogeneous elasticity.** A single β_1 for all states may miss variation by income, demographics, or culture.

Next step

A **multiple IV** approach with income controls and panel data would substantially improve credibility.

[← back to problem](#)